Topic 11: Equilibrium II

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include determining the value for an equilibrium constant for a simple esterification reaction.

Mathematical skills that could be developed in this topic include constructing expressions for K_c and K_p and calculating values with relevant units, estimating the change to the value of an equilibrium constant when a variable changes.

Within this topic, students can consider how chemists can use the concept of equilibria to predict quantitatively the direction and extent of chemical change.

Students should:	
1.	be able to deduce an expression for K_p, for homogeneous and heterogeneous systems, in terms of equilibrium partial pressures in atm
2.	be able to calculate a value, with units where appropriate, for the equilibrium constant (K_c and K_p) for homogeneous and heterogeneous reactions, from experimental data
3.	know the effect of changing temperature on the equilibrium constant (K_c and K_p), for both exothermic and endothermic reactions
4.	understand that the effect of temperature on the position of equilibrium is explained using a change in the value of the equilibrium constant
5.	understand that the value of the equilibrium constant is not affected by changes in concentration or pressure or by the addition of a catalyst